Rational-point Jacobian condition and the actual point-stalk DVR

Formal proof packet

28 October 2026

This packet records a checked pointwise differential equivalence and conditional DVR results for the actual weighted-Proj point stalk. It contains no target result, experiment, or cost ratio. The Lean source, direct compiler log, SVG graph, and CSV status table accompany this PDF.

Checked boundary

Let K be any field, $f(X) = X^5 + a_3X^3 + a_2X^2 + a_1X + a_0$, and (a, y) a rational point of $y^2 = f(a)$. The checked derivative lemma identifies the explicit $f'(a)$ used by the local-parameter proof with the formal derivative of the actual quintic. The checked equivalence is

$$(2y \neq 0 \ \vee \ f'(a) \neq 0) \quad \Longleftrightarrow \quad (\partial_y(y^2 - f(x))|_{(a,y)} \neq 0 \ \vee \ \partial_x(y^2 - f(x))|_{(a,y)} \neq 0).$$

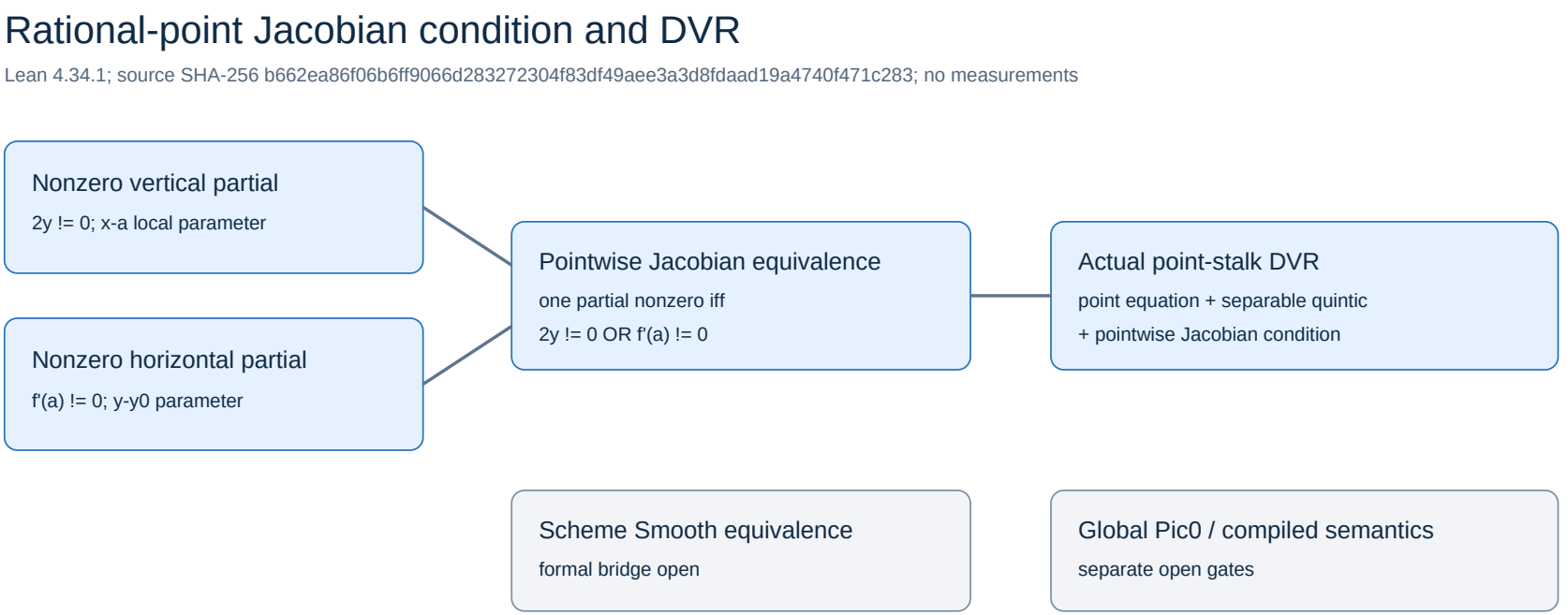
This is an algebraic equivalence between the source's explicit partial derivatives. It has no characteristic restriction. Under separability of f, either nonzero partial gives a DVR for the affine rational-point localization and the actual weighted-Proj point stalk. The two local parameters are $x - a$ in the first case and $y - y_0$ in the second. The projective theorem also accepts the nonzero-partial premise directly.

Verification and scope

Direct pinned Lean 4.34.1 compilation exited zero against Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. Five printed axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Source SHA-256:
`b662ea86f06b6ff9066d283272304f83df49aee3a3d8fdaad19a4740f471c283`.

The source combines the checked `WeightedProjNonbranchDVR` and `WeightedProjDerivativePointDVR` theorems. An equivalence with a scheme-theoretic `Smooth` predicate, global principal-divisor/Pic^0 transport, and compiled Rust semantics remain open. This result makes no order-allocation assertion in characteristic two.

Proof dependencies and open gates



Blue nodes are checked implications. Gray nodes remain open. Arrows show proof dependencies, not measured performance.